204 Force and Motion (II)

2.4.1 Whole-system and individual-body analysis

Objects constrained to have the same acceleration, for example by an inextensible string, can be treated together to relate external resultant force to total mass. Internal tensions cancel in this sum. To determine tension or contact force, then analyse an individual object's forces. Cancellation in the system equation does not mean an internal force is absent.

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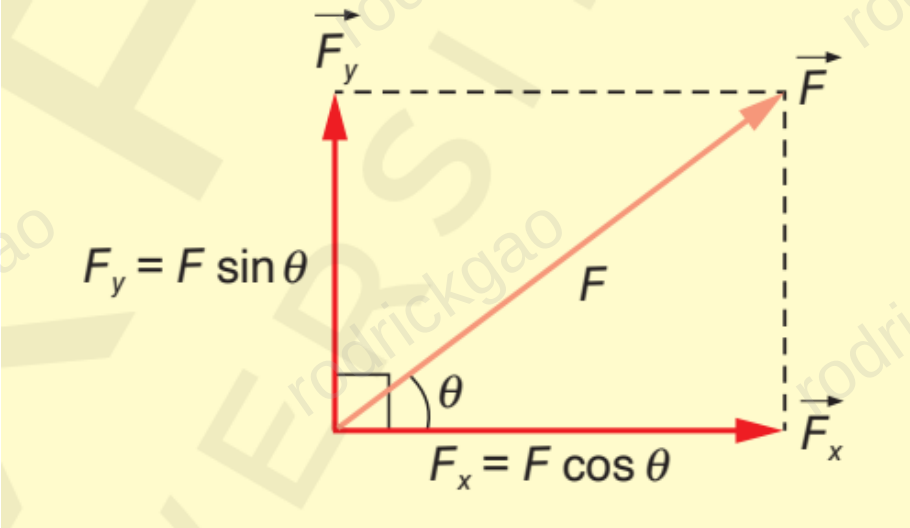
2.4.2 Resolving a force into perpendicular components

A force can be replaced by perpendicular components with the same combined effect. If its angle is measured from the positive x-axis, the adjacent component uses cosine and the opposite component uses sine. Measuring from the vertical interchanges these roles. Assign signs according to the chosen axes and each component's direction.

$$F_x = F\cos\theta$$

$$F_y = F\sin\theta$$

F: force (N); θ: angle measured from the positive x-axis to F. Components are signed.



θ is measured from the horizontal component. The adjacent component is F cos θ and the opposite component is F sin θ.

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